MEDICAL IMAGE ANALYSIS · 2025 · RESEARCH POSTER

Noise-Inspired Diffusion Model for Generalizable Low-Dose CT Reconstruction

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NEED

Train once on normal-dose data. Reconstruct low-dose CT at unseen dose levels.

NEED ranked first on every reported Mayo 2020 reconstruction and MedSAM metric.

Motivation

Lower dose changes both projection and image noise.

- A model trained for one low-dose setting may fail at another dose.
- Photon and detector noise appear in the projection. Reconstruction converts them into structured image noise.
- Using the noisy LDCT image alone to guide diffusion can copy noise or create false structures.

Question: Can models trained only on normal-dose data reconstruct doses and acquisition protocols not used during training?

Training and test inputs

TRAIN

Normal-dose CT (NDCT) projection + image

Separate models learn from NDCT projections and images.

TEST

Low-dose CT (LDCT) projection + photon count

FBP creates the LDCT image used by DGDiff.

0

low-dose training scans

Experimental setup

2016

4,825 train · 1,111 test

2020

800 external-test images

- Mayo 2016: 8 training patients, 2 held-out test patients; 50%, 25%, 12.5% and 10% dose.
- Mayo 2020: 5 chest scans at 10% and 5 abdomen scans at 25%; no retraining.

Metrics: PSNR, SSIM, RMSE, FSIM, VIF, NQM; MedSAM Dice, IoU and accuracy.

Training logic

SPDiff: shifted Poisson diffusion

Add simulated photon and electronic noise to an NDCT projection; train SPDiff to recover the clean projection.

DGDiff: doubly guided diffusion

Add Gaussian noise to an NDCT image; train DGDiff to recover the clean image.

Why two stages?

Projection noise follows photon and detector statistics. After reconstruction, the remaining image noise is more complex, so a second image model is used.

NEED: dual-domain inference pipeline

SPDiff reduces projection noise. Filtered backprojection (FBP) creates the image that DGDiff refines.

TRAIN NDCT projection → trained SPDiff + NDCT image → trained DGDiff

INFERENCE

1 LDCT

Pre-log projection

2 SPDiff

photon + detector noise reduced before FBP

Reverse projection denoising

3 FBP

$\mathcal{F}^{-1}$ after SPDiff

Projection to image

4 DGDiff

two FBP images: original LDCT, after SPDiff; guide each step

Image-domain refinement

5 NEED

Refined CT image

SPDiff

Training adds shifted-Poisson noise to normal-dose pre-log projections. At test time, SPDiff reverses this process on the measured low-dose projection before FBP.

Time-step matching

The method chooses how many reverse steps each model runs for a specific test scan. The incident photon count chooses the SPDiff starting step. After FBP, the difference between the original LDCT image and the SPDiff-FBP image chooses the DGDiff starting step. Lower dose uses more steps.

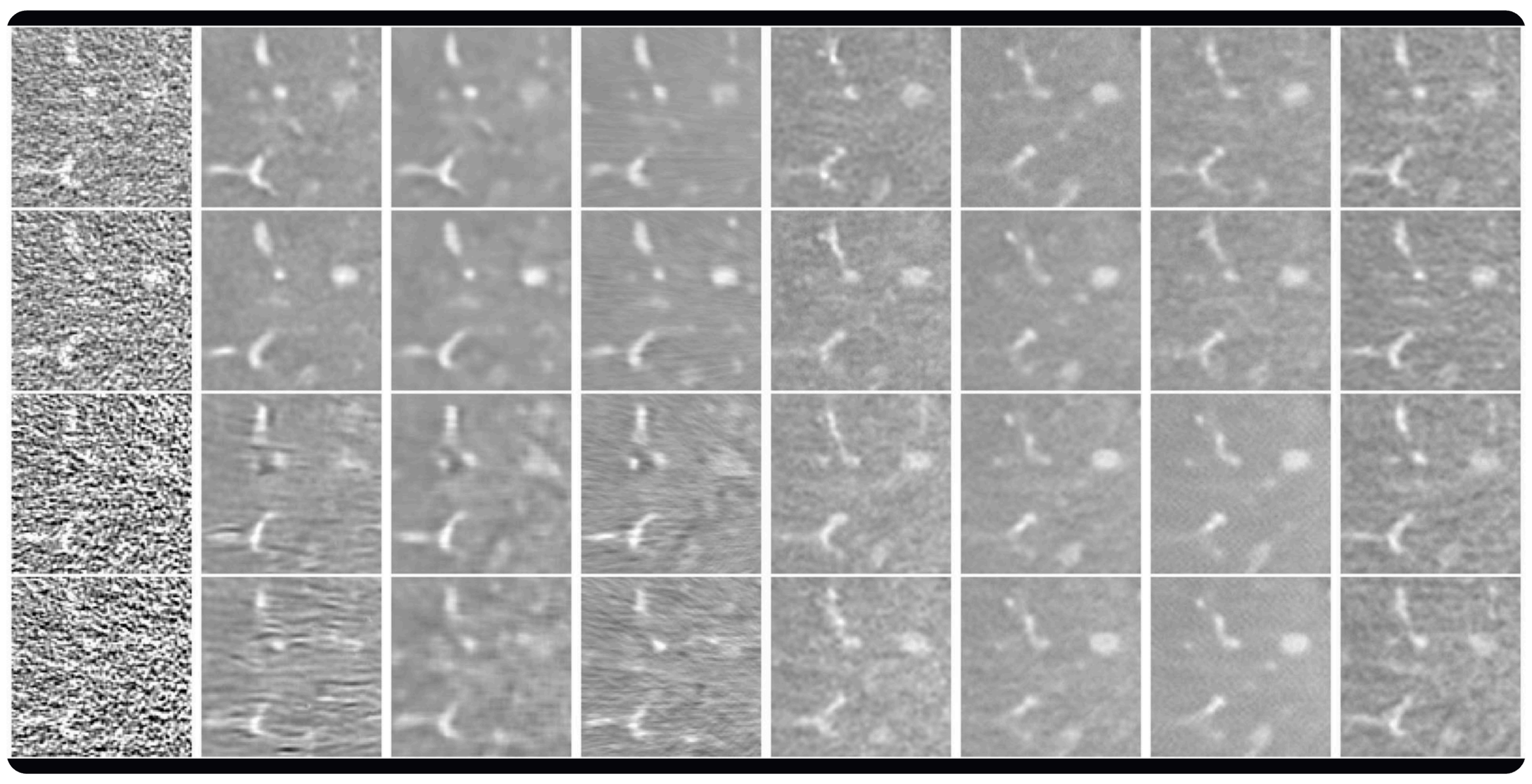
Double guidance

At each DGDiff step, the current noisy image is combined with the FBP image reconstructed directly from the original LDCT projection before the U-Net. The U-Net output is then combined with the cleaner FBP image produced after SPDiff. The first preserves structures; the second suppresses noise and artifacts.

Table 7 · 25% dose	0.121 s · matched DGDiff	15.97 s · full DGDiff	PSNR 42.82 vs 42.59 dB

Mayo 2016: seen and unseen dose levels

Figure 6 · rows: 50%, 25%, 12.5%, 10% dose · 25% is the training dose setting; the other three are unseen



Column order: NDCT · FBP · Noise2Noise · Noise2Sim · SSDDNet · DR2 · GDP · Dn-Dp · NEED

42.82 PSNR · 25%

96.75% SSIM · 25%

14.51 RMSE · 25%

Table 2, 10% dose: NEED achieved 40.90 dB PSNR, 96.07% SSIM and 18.34 HU RMSE.

Conclusion

NEED was trained without LDCT targets and achieved the best reported Mayo 2020 reconstruction and MedSAM scores. The study also tested three unseen Mayo 2016 dose levels. Validation on real raw projections from multiple scanners and centers is still missing.

- Table 5: best PSNR, SSIM, RMSE
- Table 6: best Dice, IoU, accuracy
- Table 8: double guidance improves all metrics

Mayo 2020 reconstruction

Unseen Mayo 2020 dataset · chest 10%, abdomen 25% · Table 5

37.65 PSNR (dB)

90.30 SSIM (%)

29.72 RMSE (HU)

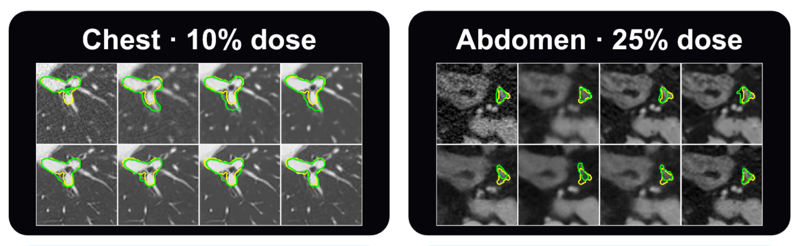
vs best prior method: +1.15 dB · +1.22 pp · −12.7%

PSNR ↑	Noise2Sim	36.50
	SPDiff	36.78
	NEED	**37.65**
SSIM ↑	GDP	89.08%
	SPDiff	88.42%
	NEED	**90.30%**
RMSE ↓	Noise2Sim	34.04
	SPDiff	33.09
	NEED	**29.72**

Source: paper Table 5. Values are mean ± std; headline values show the mean.

Mayo 2020 segmentation

Fig. 9 · MedSAM masks: yellow NDCT reference, green reconstruction



Dice 0.924 IoU 0.859 · Acc 99.9572%

Dice 0.845 IoU 0.732 · Acc 99.9832%

Table 6: NEED ranked first for Dice, IoU and accuracy in both groups.

Discussion

Ablation: single vs double guidance

+1.63 PSNR (dB)

+1.25 SSIM (pp)

−3.16 RMSE (HU)

With SPDiff fixed, double guidance improved every Table 8 metric over single guidance at 25% dose.

Limits of the evidence

- At 12.5% and 10% dose, the paper reports vessel blurring or artifacts; performance remains dose-dependent.
- Mayo 2016 uses simulated fan-beam projections. Mayo 2020 projections were generated from LDCT images, not raw scanner data.
- These tests omit scatter, detector nonlinearity and other real pre-log effects.
- Segmentation uses MedSAM masks from NDCT as references, not manual expert annotations.
- Multi-center and multi-device validation is still needed; the 512^2 stage also exceeds 0.5 s per slice.

Computational cost

75.76M parameters

11 h SPDiff training

48 h DGDiff 256^2

117 h super-resolution

One RTX 4090 (24 GB). 256^2 inference: 0.205 s at 25%; 0.250 s at 10%.

REFERENCE

Gao Q, Chen Z, Zeng D, et al. Noise-Inspired Diffusion Model for Generalizable Low-Dose CT Reconstruction. Medical Image Analysis. 2025;105:103710.

Figure sources: paper Figures 6 and 9. Metrics: Tables 2, 4, 5, 6, 7 and 8.

PAPER

arxiv.org/abs/2506.22012

CODE

github.com/qgao21/NEED

METHOD

NDCT-only training
scan-specific denoising steps